Reviewer Pack — Minimal Rank of Geometric Langlands Duality over Function Fi…

Entry 987c3b20d2ce · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Geometric Langlands Duality over Function Fields vs AC0 Parity Circuit Size

Entry ID: `987c3b20d2ce` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-25 10:51:09 UTC

1. Conjecture

Field A (mathematical branch): Geometric Langlands Theory **Field B** (complexity object): Complexity Theory: AC⁰ Parity Circuit Complexity

Statement:

> {'s1': 'For any function field K with genus g ≥ 2, there exists a polynomial-time computable invariant ψ_K(C) associated with an AC0 parity circuit C over K such that ψ_K(C) = Θ(log^3(|C|/g)) for all circuits C.', 's2': 'The polynomial-time computable invariant satisfies ψ_K(C) > 1 for any AC0 parity circuit C computing the function f: {0,1}^n -> {0,1} with f being a non-trivial Boolean function.', 's3': "Conversely, if there exists an AC0 parity circuit C with ψ_K(C) < 1, then there exists a field K' with genus g' ≤ g such that C can be naturally interpreted as an AC0 parity circuit over K'."}

Rationale (proposer's reasoning):

> {'s1': 'Geometric Langlands duality offers a bridge between geometry and number theory that may provide new insights into the complexity of Boolean functions.', 's2': 'By leveraging geometric invariants, we could potentially establish non-trivial lower bounds on AC0 parity circuit size, which is a fundamental problem in computational complexity theory.', 's3': 'The proposed invariant ψ_K(C) would provide a new method for studying AC0 parity circuits that is both natural and powerful.'}

Taxonomy category: `AC0_PARITY` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `9897294d2d11dd51`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> For each AC0 parity circuit C, the invariant ψ_K(C) must be within Θ(log^3(|C|/g)) where g is the genus of the function field K associated with C.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM₁	LLM₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - geometric langlands duality AND ac0 parity circuit size - polynomial-time computable invariant ψ_K(C) AND geometric langlands theory - AC0 parity circuit complexity AND minimal rank of function fields

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.1s

5.1 Generated Python source

```python
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_random_boolean_function(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def construct_function_field(g):
        # Simplified construction of a function field with genus g
        return f"K_{g}"

    def compute_invariant(C, K):
        n = len(C)
        g = int(K[2:])
        if n == 0 or g == 0:
            return None
        return (math.log(n) / math.log(2)) ** 3 / g

    def check_conjecture(C, K):
        psi_K_C = compute_invariant(C, K)
        if psi_K_C is None:
            return False, "mapping_undefined"
        expected_range = (math.log(len(C)) / math.log(2)) ** 3 / int(K[2:])
        return psi_K_C >= expected_range and psi_K_C <= expected_range * 1.05

    n_values = [5, 10, 15, 20, 30, 40]
    instances_tested = 0
    conjecture_holds = True
```

```
    counterexample = ""

    for n in n_values:
        f = generate_random_boolean_function(n)
        K = construct_function_field(2)  # Simplified genus g=2
        C = f  # Simplified AC0 parity circuit
        instances_tested += 1

        holds, desc = check_conjecture(C, K)
        if not holds:
            conjecture_holds = False
            counterexample = desc

    return {
        "metric_name": "conjecture_support",
        "metric_value": instances_tested,
        "instances_tested": instances_tested,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or list(range(2, 50))  # Default to first 30 primes

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(result
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_frac
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_frac
    else:
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"mapping_undefined\" first_failing_seed={first_f
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e68d0804.py", line 72, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e68d0804.py", line 53, in run_trial
```

```
    holds, desc = check_conjecture(C, K)
    ^^^^^^^^^^^
TypeError: cannot unpack non-iterable bool object
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

> The test code crashed before producing data, which prevents us from verifying whether the invariant ψ_K(C) meets the pre-registered support condition. | next: Investigate and fix the crash in the test code to verify the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	12076
2	propose	ollama_remote	glm4:latest	0	0	12366
3	preregistration	ollama_remote	glm4:latest	0	0	5439
4	novelty	ollama_remote	glm4:latest	0	0	4649
5	novelty	ollama_remote	glm4:latest	0	0	5577
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16566
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8612
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13866
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8328
10	judge	ollama_remote	glm4:latest	0	0	21438

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 108917 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/987c3b20d2ce.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/987c3b20d2ce.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/987c3b20d2ce.tar.gz` (if generated)